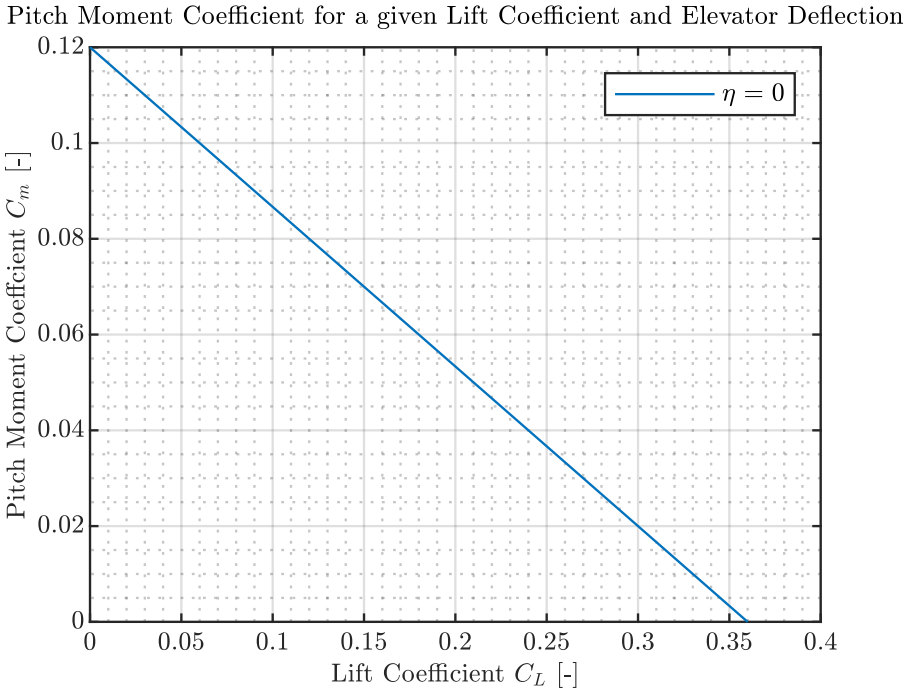
Exercise 8: Trim

A stable aircraft configuration shows the following characteristics:

Lift coefficient at zero AoA	C_{L0}	0,3
Pitch moment coefficient at zero AoA	C_{m0}	0,02
Elevator pitch moment derivative	$C_{m\eta}$	+-2.5
Elevator lift derivative	$C_{L\eta}$	+-0.5
Aircraft mass	m	2000 kg
Altitude	h	3000 m
Airspeed in Mach	M	0,2
Reference area	S_{ref}	20 m^2

Note: at this point we don't know the sign of the derivatives. That's why their sign is denoted by „+-". The following relation between the lift- and the moment-coefficient (for one particular elevator deflection!) is known:



a. What is the correct sign of $C_{m\eta}$?
b. How does the relation/picture (qualitatively) change when the pilot pulls/pushes the stick?
c. What difference does the curve „**Lift coefficient over angle of attack**" show, if instead of keeping the elevator constant, the elevator is always moved to the **trim condition**? Derive this effect from the equations in the lecture slides and give signs of the involved derivatives.
d. Calculate the required elevator deflection for a steady-state horizonal cruise flight. Hint: You can assume the standard atmosphere.

Additional data:

Density at MSL according to ISA standard atmosphere:	$\rho_{s,MSL} = 1{,}225 \frac{kg}{m^3}$
Pressure at MSL according to ISA standard atmosphere:	$p_{s,MSL} = 1{,}01325e5\ Pa$
Temperature at MSL according to ISA standard atmosphere:	$T_{s,MSL} = 288{,}15\ K$
Gravitational acceleration:	$g_0 = 9{,}80665 \frac{m}{s^2}$
Specific gas constant:	$R = 287{,}05 \frac{J}{kg\ K}$
Polytropic exponent:	$n = 1{,}235$
Isentropic exponent:	$\kappa = 1{,}4$
Radius of the earth:	$r_{earth} = 6356766\ \mathrm{m}$